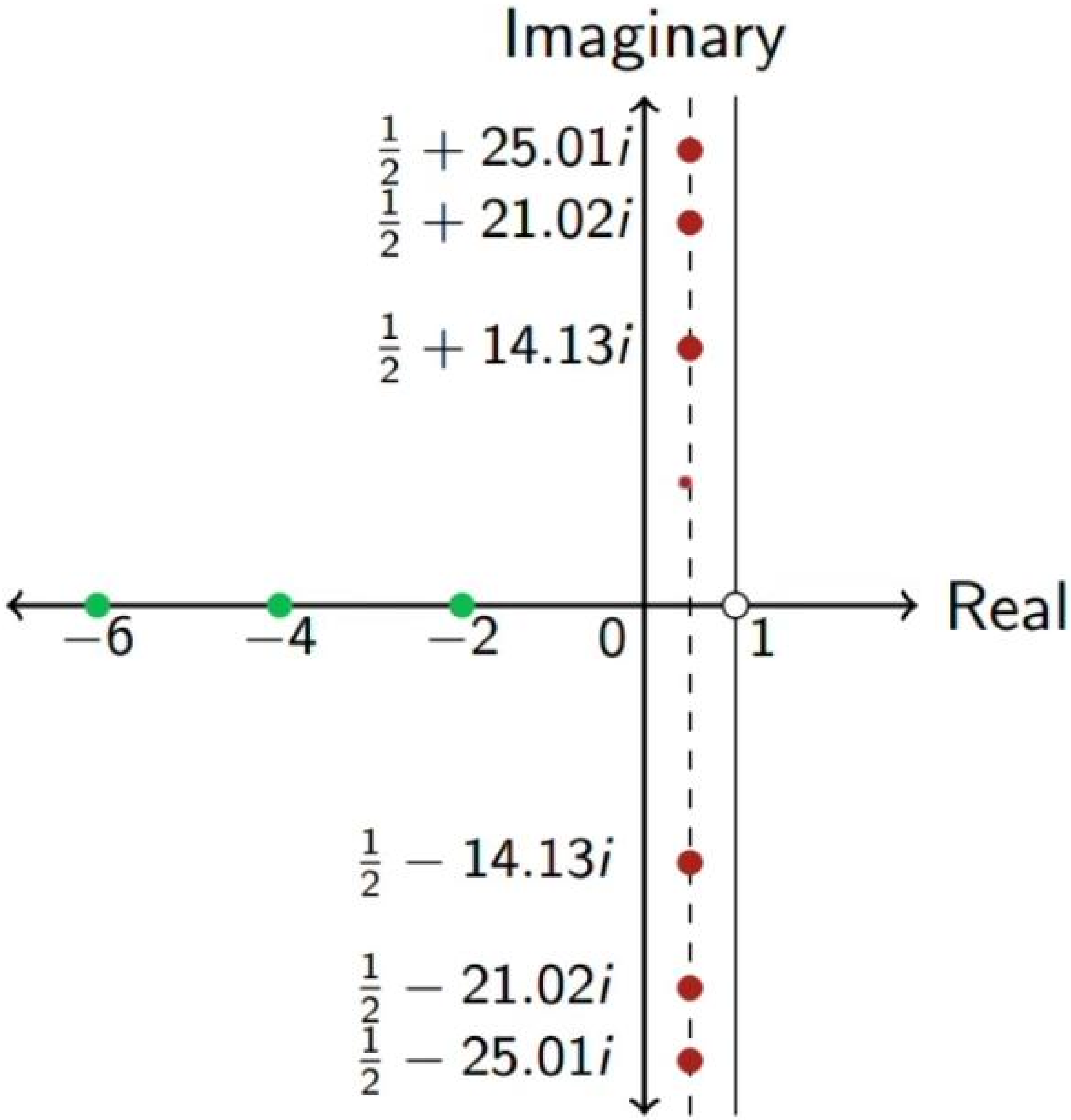
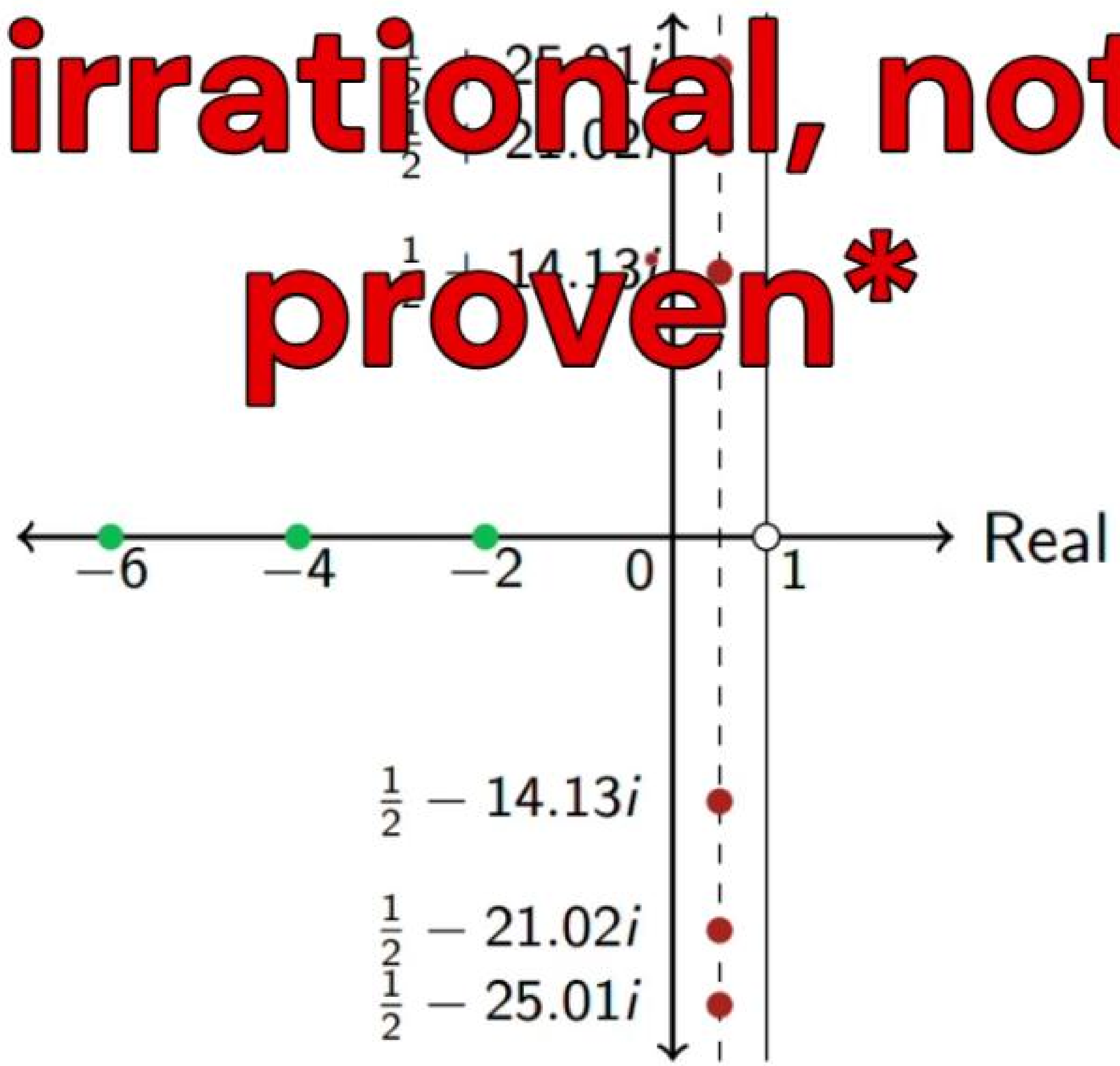
Zeta in the Complex Plane



- $\zeta(s)$ is a complex function.
- It has a pole at $s = 1$
- An interesting question is when does $\zeta(s) = 0$?
- It has trivial zeros at negative even integers, and an infinite number of nontrivial zeros in the "critical strip" where $0 < Re(s) < 1$.
- Riemann Hypothesis (1859): All of the nontrivial zeros are on the "critical line" $Re(s) = \frac{1}{2}$
- \$1 million for proof or disproof

Zeta in the Complex Plane

believed to be irrational, not proven*



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Scope

- The Basel problem: $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$
- Following the first chapter of *The Riemann Zeta-Function* by Aleksandar Ivić (1985)
  - The Euler product formula
  - Analytic continuation
  - Laurent series for $\zeta(s)$ at $s = 1$
  - The functional equation relating $\zeta(s)$ to $\zeta(1-s)$
  - The Hadamard product formula
  - The Riemann-Von Mangoldt equation
  - A statement of the Riemann Hypothesis and the Lindelöf Hypothesis
  - (Don't worry if you have no idea what any of these mean–the point of the video series it to explain them!)

Prerequisites

- Calculus
- You definitely need to know calculus!
- A lot of proofs will involve integrals and infinite series–so you better know calculus!
- Complex numbers ($z = a + bi$, where $i = \sqrt{-1}$)
- It will certainly help if you know complex analysis, which is the topic of this series. But I'll try to explain things you need to know.

The Zeta Function

- The Riemann zeta function is defined for real numbers $s > 1$ as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \cdots .$$

- $\zeta(2) = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$, a.k.a. the Basel Problem
- $\zeta(1) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$, a.k.a. the harmonic series, which diverges.
- Plugging in any $s < 1$ to the right-hand-side also diverges, as each term is at least as big as the corresponding term for $s = 1$.
- Eventually, we'll extend $\zeta(s)$ to a complex function for all complex values of s other than $s = 1$.
- In this extended function, $\zeta(-1) = -\frac{1}{12}$. Recall an infamous Numberphile video (2014) where they claimed

$$1 + 2 + 3 + 4 + \cdots = -\frac{1}{12}.$$

The Basel Problem

- Mengoli (1650):

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = ???$$

- Euler (1734-5):

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

- His proof is not considered rigorous by modern standards, but it's mostly right and has some deep insights.

Numerical Approximation

- Before proving it, Euler numerically estimated what the answer should be.
- Adding up the partial sums of $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$ is too slow–the first 10,000 terms summed together gets you 1.644834..., which is correct to only 3 decimal places!
- So he used a summation technique known as Euler-Maclaurin summation, where he got 20 decimal places of accuracy: 1.64493406684822643647...
- By 1730, π had been calculated to 100+ digits
- Aha! That's $\frac{\pi^2}{6}$ (???)

Euler's proof of the Basel Problem

Taylor series of $\sin x$:

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$$

$$\frac{\sin x}{x} = 1 - \frac{1}{3!}x^2 + \frac{1}{5!}x^4 - \cdots$$

Euler assumed he could factor $\sin x$ as a product of terms involving its zeros. Then he showed the first terms are:

$$\frac{\sin x}{x} = 1 - \frac{\zeta(2)}{\pi^2}x^2 + \cdots$$

The x^2 term on the left must equal the x^2 term on the right, so

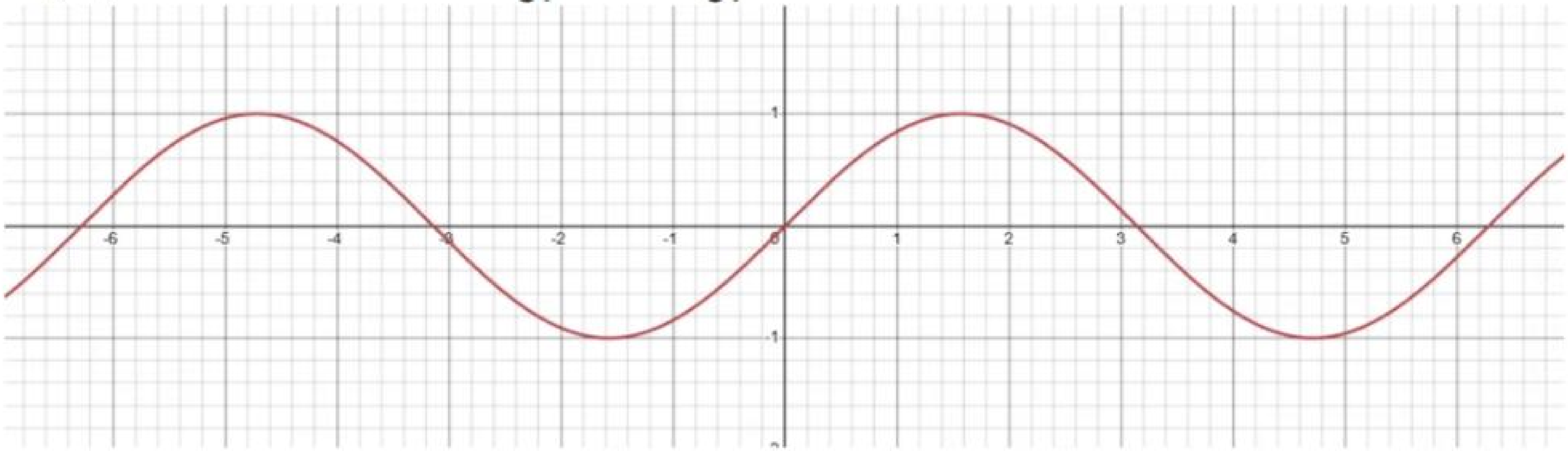
$$-\frac{1}{3!} = -\frac{\zeta(2)}{\pi^2}$$

Therefore,

$$\zeta(2) = \frac{\pi^2}{6}$$

Euler's proof of the Basel Problem

$f(x) = \sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$



$\sin x = 0$ if $x = \pi n x$, for all integers n

Note the slope at x=0 is 1 ($f'(0) = \cos(0) = 1$)

sin x = 0 if x = πn, not x = πnx*

Euler's proof of the Basel Problem

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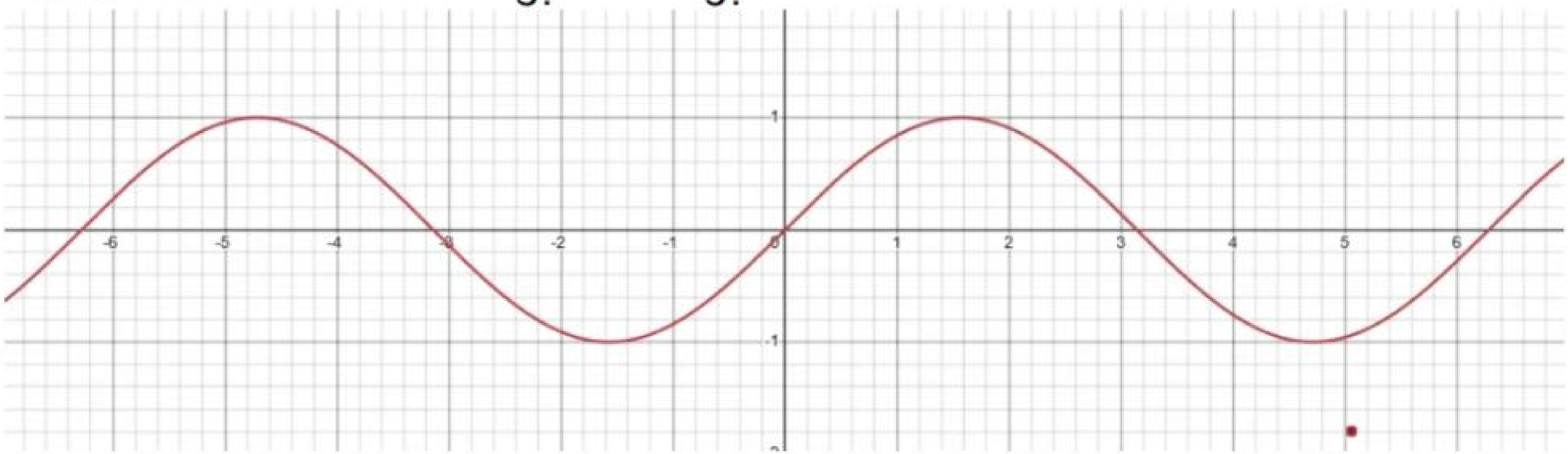

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Any series of $\sin x$ must start with $x + \cdots$, not e.g. $24x + \cdots$

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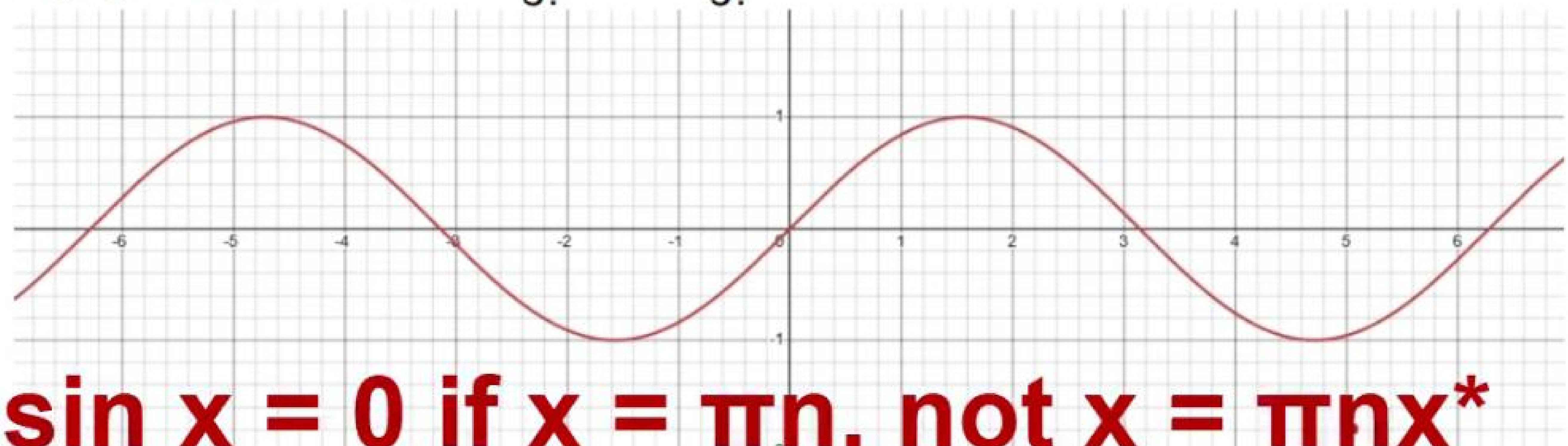

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Euler's proof of the Basel Problem

- x
- $x(x-1)$
- $x(x-1)(x-2)$
- $x(x-1)(x-2)(x-3)$
- $x(x-1)(x-2)(x-3)(x-4)$

- x
- $-x+x^2$
- $2x-3x^2+x^3$
- $-6x+11x^2-6x^3+x^4$
- $24x-50x^2+35x^3-10x^4+x^5$

The coefficient of x is growing like $n!$

Factorial

For positive integers n, the factorial is defined as

$$n! = n(n-1)\cdots 2 \times 1$$

- $1! = 1$
- $2! = 2 \times 1 = 2$
- $3! = 3 \times 2 \times 1 = 6$
- $4! = 4 \times 3 \times 2 \times 1 = 24$

This satisfies the functional equation $n! = n \times (n-1)!$ Using this, we see that $1! = 1 \times 0!$, which implies $0! = 1$.

We'll eventually need to know the "factorial" of a non-integer or complex number!

Euler's proof of the Basel Problem

- $\frac{1}{0!}x$
- $\frac{1}{1!}x(x-1)$
- $\frac{1}{2!}x(x-1)(x-2)$
- $\frac{1}{3!}x(x-1)(x-2)(x-3)$
- $\frac{1}{4!}x(x-1)(x-2)(x-3)(x-4)$

- $\frac{1}{0!}(x)$
- $\frac{1}{1!}(-x+\cdots)$
- $\frac{1}{2!}(2x+\cdots)$
- $\frac{1}{3!}(-6x+\cdots)$
- $\frac{1}{4!}(24x+\cdots)$

Euler's proof of the Basel Problem

- $\frac{1}{0!}x$
- $\frac{1}{1!}x(x-1)$
- $\frac{1}{2!}x(x-1)(x-2)$
- $\frac{1}{3!}x(x-1)(x-2)(x-3)$
- $\frac{1}{4!}x(x-1)(x-2)(x-3)(x-4)$

- x
- $-x+\cdots$
- $x+\cdots$
- $-x+\cdots$
- $x+\cdots$

Euler's proof of the Basel Problem

$$
\begin{aligned}
x &= x \\
x(x-1) &= -x(1-x) \\
x(x-1)(x-2) &= x(1-x)(2-x) \\
x(x-1)(x-2)(x-3) &= -x(1-x)(2-x)(3-x) \\
x(x-1)(x-2)(x-3)(x-4) &= x(1-x)(2-x)(3-x)(4-x)
\end{aligned}
$$

Euler's proof of the Basel Problem

- $\frac{1}{0!}x$
- $\frac{1}{1!}x(1-x)$
- $\frac{1}{2!}x(1-x)(2-x)$
- $\frac{1}{3!}x(1-x)(2-x)(3-x)$
- $\frac{1}{4!}x(1-x)(2-x)(3-x)(4-x)$

- x
- $x+\cdots$
- $x+\cdots$
- $x+\cdots$
- $x+\cdots$

Euler's proof of the Basel Problem

$$
\begin{aligned}
&\frac{1}{4!}x(1-x)(2-x)(3-x)(4-x)\\
=&\frac{1}{1}\cdot\frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{4}\cdot x(1-x)(2-x)(3-x)(4-x)\\
=&x\cdot\frac{1}{1}(1-x)\cdot\frac{1}{2}(2-x)\cdot\frac{1}{3}(3-x)\cdot\frac{1}{4}(4-x)\\
=&x(1-x)\left(1-\frac{x}{2}\right)\left(1-\frac{x}{3}\right)\left(1-\frac{x}{4}\right)
\end{aligned}
$$

- Remember this is a function with zeros at $0, 1, 2, 3, 4$, and the first term of its Taylor series is x (as opposed to e.g. $24x$).

Euler's proof of the Basel Problem

$$x(1-x)\left(1-\frac{x}{2}\right)\left(1-\frac{x}{3}\right)\left(1-\frac{x}{4}\right)$$

- Remember this is the factorization of a function with zeros at $0, 1, 2, 3, 4$, and the first term of its Taylor series is x (as opposed to e.g. $24x$).
- $\sin x$ has zeros at $0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$, and the first term of its Taylor series is x. Euler concluded that its factorization must be

$$x\left(1-\frac{x}{\pi}\right)\left(1+\frac{x}{\pi}\right)\left(1-\frac{x}{2\pi}\right)\left(1+\frac{x}{2\pi}\right)\left(1-\frac{x}{3\pi}\right)\left(1+\frac{x}{3\pi}\right)\cdots$$

- This step is sketchy because you can't just apply a result that works for finite cases to infinite cases. It was proved much later that you can indeed factor functions into infinite products based on their zeros–the Weierstrass Factorization Theorem (1870s).

Euler's proof of the Basel Problem

$$x\left(1-\frac{x}{\pi}\right)\left(1+\frac{x}{\pi}\right)\left(1-\frac{x}{2\pi}\right)\left(1+\frac{x}{2\pi}\right)\left(1-\frac{x}{3\pi}\right)\left(1+\frac{x}{3\pi}\right)\cdots$$

- Remember that $(a-b)(a+b)=a^2-b^2$. So each pair of terms can be combined, for example:

$$\left(1-\frac{x}{3\pi}\right)\left(1+\frac{x}{3\pi}\right)=1-\frac{x^2}{9\pi^2}$$

- So the factorization becomes

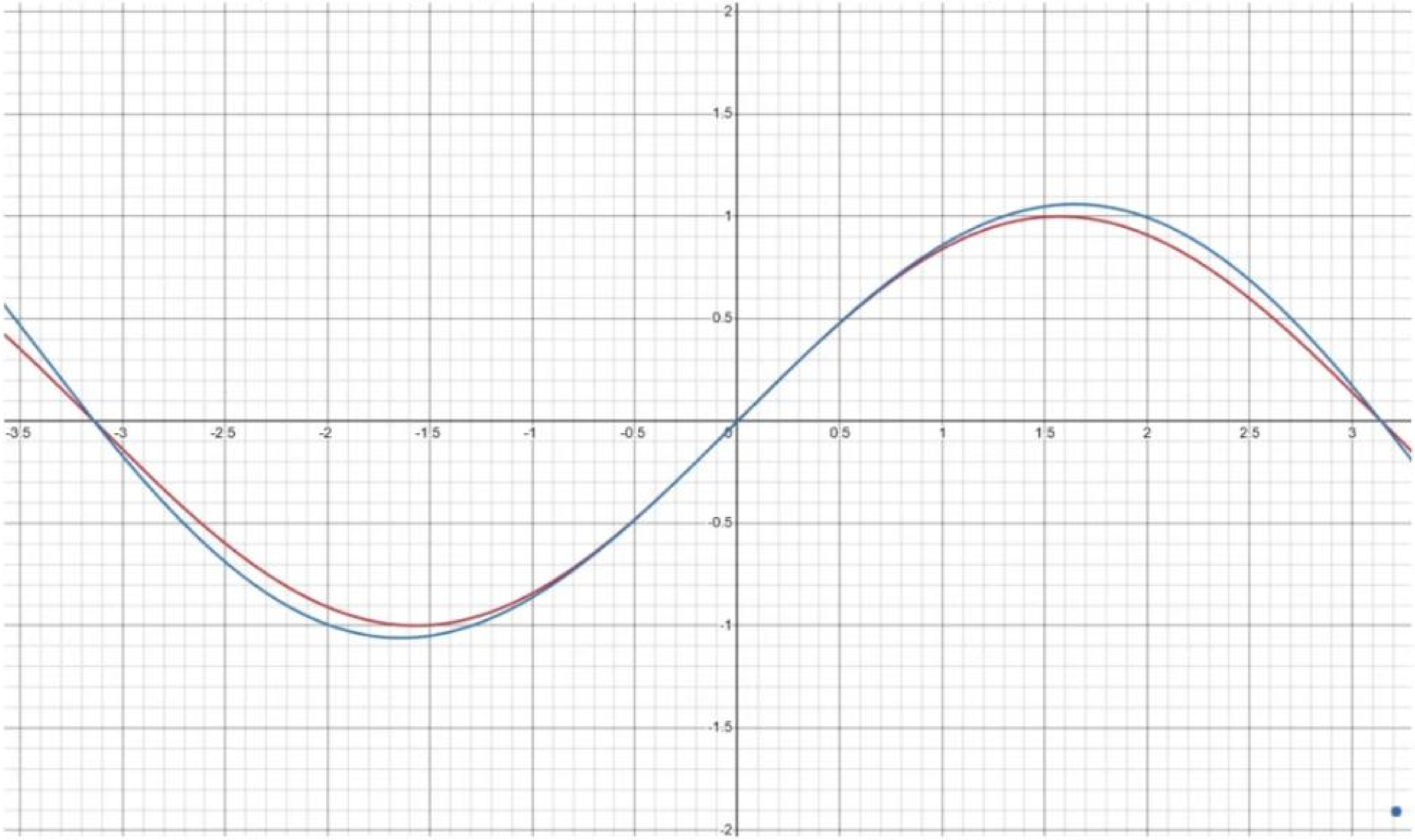
$$\sin x = x\left(1-\frac{x^2}{\pi^2}\right)\left(1-\frac{x^2}{4\pi^2}\right)\left(1-\frac{x^2}{9\pi^2}\right)\cdots$$

- Dividing both sides by x:

$$\frac{\sin x}{x}=\left(1-\frac{x^2}{\pi^2}\right)\left(1-\frac{x^2}{4\pi^2}\right)\left(1-\frac{x^2}{9\pi^2}\right)\cdots$$

Euler's proof of the Basel Problem

$$f(x) = \sin x$$
$$g(x) = x\left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\left(1 - \frac{x^2}{16\pi^2}\right)$$



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$f(x) = \sin x$

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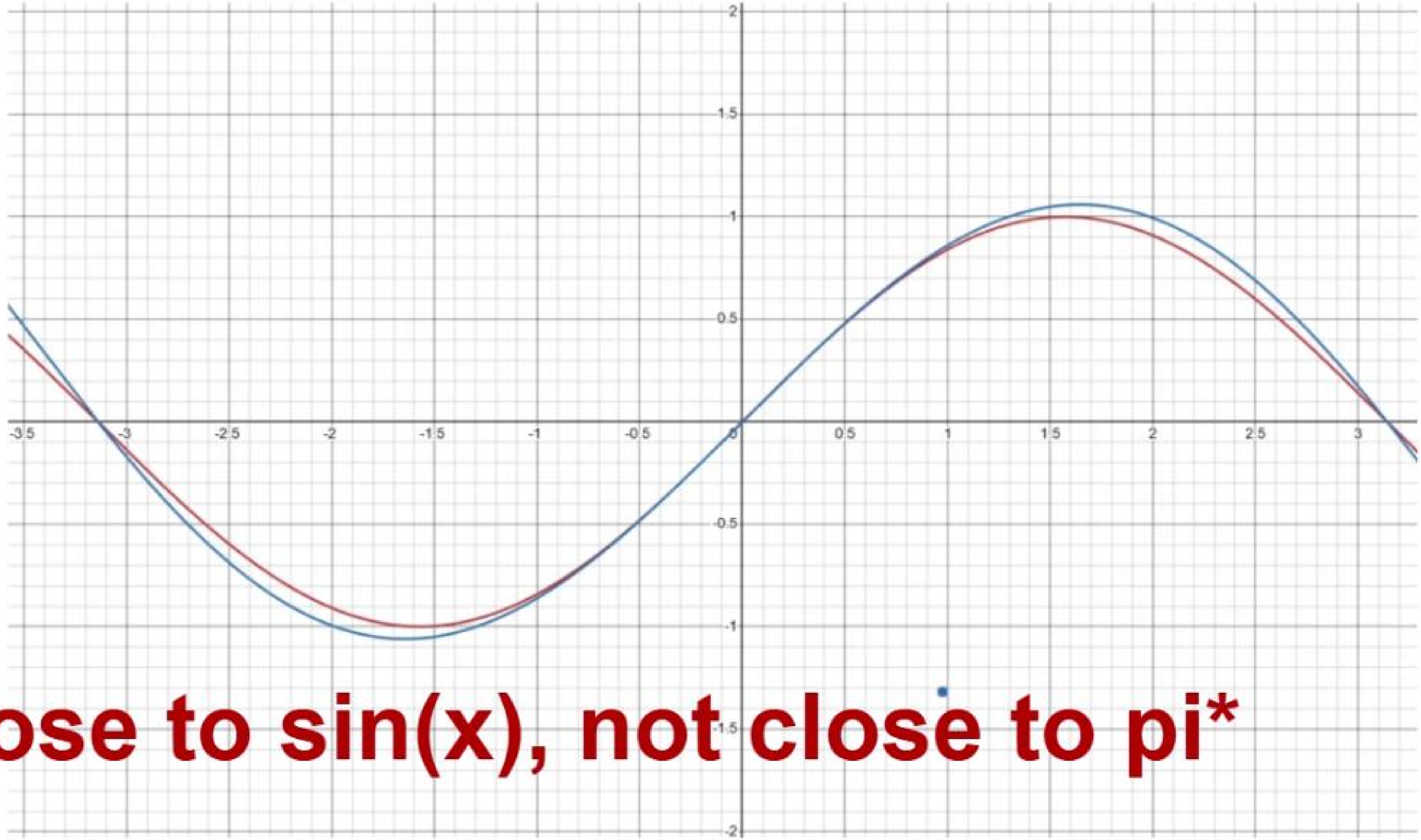

close to sin(x), not close to pi*

Euler's proof of the Basel Problem

$$\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\cdots$$

- Multiply out the product and collect terms carefully:
- All the 1's multiplied out becomes 1. So

$$\frac{\sin x}{x} = 1 + \cdots$$

- There can't be an x term, so the next term has to be x^2. The x^2 term involves selecting a 1 from every term in the product except one term, where we select the $-\frac{x^2}{n^2\pi^2}$ term. Hence

$$\begin{aligned}\frac{\sin x}{x} &= 1 + \left(-\frac{1}{\pi^2} - \frac{1}{4\pi^2} - \frac{1}{9\pi^2} - \cdots\right)x^2 + \cdots \\ &= 1 - \frac{1}{\pi^2}\left(1 + \frac{1}{4} + \frac{1}{9} + \cdots\right)x^2 + \cdots \\ &= 1 - \frac{1}{\pi^2}\zeta(2)x^2 + \cdots\end{aligned}$$

Euler's proof of the Basel Problem

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Therefore,

$$\zeta(2) = \frac{\pi^2}{6}$$

Euler's proof of the Basel Problem

- Next time: Analyzing $\zeta(s)$ in the complex plane and discussing convergence.

Feedback

- Let me know if you have feedback!
- Was the pacing too fast/too slow?
- Was subject too easy/too difficult?
- Did I skip over important steps?
- Were there parts that didn't make sense or seemed unjustified?
- Was it interesting or did it seem pointless?
- Did you spot any errors? I probably made some.
- What can I do better?